

- Q.29 Write five differences between CNC & DNC machine.
- Q.30 Write a short note on Canned cycles.
- Q.31 Name any five M codes.
- Q.32 Write advantages of Robots.
- Q.33 Explain the concept of group technology.
- Q.34 What are the Common problem in Mechanical parts of a CNC?
- Q.35 Explain advantages and disadvantages of automation.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain construction and working of LVDT with neat sketch.
- Q.37 Explain CNC machines. Write its advantages, disadvantages and applications.
- Q.38 Write short note on :
- FMS
 - Touch Sensor

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4th Sem / Mech. Engg. (MSIL) Subject:- CNC Machines and Automation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The full form of MCU is
- Machine computer unit
 - Machine Control universal
 - Machine Control unit
 - Machine Computer universal
- Q.2 A symmetrical layout to accommodate the required number of tool is called
- ATC
 - Actuator
 - Punch Tape & Tool
 - Tool magazines
- Q.3 The open loop control system is also Called
- Non-feedback control system
 - Feedback control system
 - Mechanical Control system
 - None of the above
- Q.4 Which sensors are used to detect the presence or an absence of an object.

- a) Tachometer b) Actuator
- c) LVDT d) Opto Interrupter
- Q.5 Which of the following code is use for spindle stop ?
- a) M00 b) M03
- c) M04 d) M05
- Q.6 Which of the following code is use for Thread cutting cycle.
- a) G 32 b) G 40
- c) G 90 d) G 92
- Q.7 Which of the following is not a part of robot.
- a) Transducers b) Actuators
- c) Controller d) Sensors
- Q.8 A set of instructions, inbuilt or stored in the system memory, to perform a fixed sequence of operation is called
- a) Absolute system b) Canned cycles
- c) Do Loops d) Tool offsets
- Q.9 M10 Codes used for
- a) Chuck close b) Spindle stop
- c) Chuck open d) Coolant ON
- Q.10 Several machine tools can be controlled by a central computer in
- a) NC b) DNC
- c) CNC d) CCNC

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define NC.
- Q.12 Write components of CNC Machines.
- Q.13 Define ATC.
- Q.14 Define Actuators.
- Q.15 Write full form of LVDT.
- Q.16 What is M-code?
- Q.17 Define CAD.
- Q.18 What is Automation.
- Q.19 What do you mean by EOB?
- Q.20 Define Transducer.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write advantages of CNC machines.
- Q.22 Explain Machine control unit.
- Q.23 Write a short note on feedback system.
- Q.24 Explain swarf Removal & its methods.
- Q.25 Write various specification of CNC machines.
- Q.26 Explain open loop and closed loop system.
- Q.27 Write short note on opto-interrupters.
- Q.28 Write short note on servomotors.